REPRODUCIBILITY AND ENHANCEMENT OF DEEP LEARNING MODELS FOR THE SEN2FIRE DATASET

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***Abstract*—Rapid and accurate wildfire detection is critical for mitigating the catastrophic environmental and economic impacts of uncontrolled fires. The Sen2Fire dataset facilitates research in this domain by combining Sentinel-2 multispectral imagery with Sentinel-5P aerosol products to address challenges such as smoke occlusion. In this work, we conduct a comprehensive re-evaluation of the Sen2Fire baselines and establish a new state-of-the-art performance standard. Furthermore, we employ advanced segmentation architectures, including nnU-Net and Transformer-based models (SegFormer) and assess the impact of Test Time Augmentation (TTA). Our experiments demonstrate that the nnU-Net framework establishes a new state-of-the-art on this dataset, achieving an F1-score of 0.3675 compared to the original baseline of 0.2810. Furthermore, contrary to the original study's conclusion that selecting specific band subsets yields superior performance, our results indicate that robust architectures effectively leverage the full 13-band multispectral input to maximize detection accuracy. This study provides a rigorous re-evaluation of the Sen2Fire benchmark and offers a strong baseline for future research in satellite-based wildfire monitoring.**

***Index Terms*—Wildfire detection, Remote sensing, Deep learning, Semantic segmentation, Multispectral imaging, Sentinel-2, nnU-Net.**

I. INTRODUCTION

Wildfires pose an increasing threat to global ecosystems and economies [1, 2]. As evidenced by the catastrophic 2019–2020 Australian bushfire season [3], rapid and accurate detection of these events is critical for disaster response. While satellite imagery offers a scalable solution for monitoring, it presents significant challenges [4], particularly regarding the high class-imbalance between fire and non-fire pixels and the complexity of distinguishing smoke plumes from cloud cover.

The primary objective of this study is to rigorously evaluate and surpass the deep learning baselines established for the Sen2Fire dataset [5]. The original study proposed a standard U-Net architecture [6] and suggested that utilizing subsets of spectral bands yielded superior performance compared to using the full spectral range.

In this work, we validate these claims through reproducibility and further surpass the baseline by introducing state-of-the-art approaches, including the nnU-Net framework [7] and Transformer-based models such as SegFormer [8]. We first replicate the original U-Net baseline to ensure a fair comparison. We then show that the automatically configured nnU-Net produces an improved U-Net variant tailored to our dataset, significantly outperforming previous methods and setting a new state-of-the-art benchmark. Crucially, our experiments challenge the prior hypothesis regarding spectral selection; we provide evidence that robust architectures can effectively leverage the full 13-band multispectral input to maximize detection accuracy, rather than relying on band subsets. Finally, we perform a comparative analysis between CNN-based approaches and Transformer variants (e.g., Mixed-TransformersCNN [9]) to evaluate their efficacy in capturing large-scale smoke features.

II. SEN2FIRE DATASET DESCRIPTION

The Sen2Fire is a satellite-based wildfire detection dataset derived from Sentinel-2 MSI (MultiSpectral Instrument) [10] and Sentinel-5P aerosol product [11]. It is designed for pixel-level segmentation of wildfire-affected areas and for benchmarking deep learning models for burned-area mapping and early fire detection.

A. Data Structure

As detailed in Table II, each sample is stored as a 512×512 patch in an `.npz` file containing three arrays: `image` $(12, 512, 512)$, comprising 12 resampled Sentinel-2 bands (B1–B12); `aerosol` $(512, 512)$, representing band B13; and a binary `label` $(512, 512)$ where 1 denotes fire. For training, the spectral and aerosol arrays are concatenated into a 13-channel tensor to support band-selection strategies such as *RGB+aerosol*, *SWIR+aerosol*, or *vanilla+aerosol* (all 13 channels).

B. Spectral Bands

The dataset covers all primary Sentinel-2 MSI bands relevant to vegetation, surface reflectance and fire analysis. Key bands include NIR, SWIR and Red, which are highly responsive to burned surfaces and vegetation stress.

C. Dataset Splitting

The dataset is divided into three subsets with no overlap:

- **Training**: 1458 patches
- **Validation**: 504 patches
- **Test**: 504 patches

As we can see in Table I all the data subsets are highly imbalanced, with fire pixels representing only a small fraction of the total. This strong imbalance reflects the natural prevalence of non-fire areas in satellite imagery and highlight the importance of robust training strategies to mitigate bias toward the majority class.

III. Preprocessing and Data Augmentation

All Sen2Fire samples (across training, validation and test splits) undergo the same preprocessing steps defined in this section.

A. Preprocessing Pipeline

The preprocessing workflow consists of three primary steps: band construction, dataset statistics calculation and input normalization.

1) Band Construction and Input Strategies: Each sample in the Sen2Fire dataset consists of 12 Sentinel-2 spectral bands, a Sentinel-5P aerosol index and a binary segmentation mask. Consistent with the original Sen2Fire study, which demonstrated that selecting specific band subsets yields superior performance to using all bands, we limited our experiments to two cases:

- **Vanilla + Aerosol**: Utilizing all 13 available bands (12 spectral + 1 aerosol).
- **SWIR + Aerosol**: Combining bands B4, B8, B12 and the aerosol index (B13). We selected this configuration as it was reported as the top-performing input combination in the original benchmark.

2) Dataset Statistics Calculation: To ensure correct normalization and prevent data leakage, the dataset mean and standard deviation were computed using only the training split. For each spectral channel, we computed:

$$\mu = \frac{\sum x}{N}, \qquad \sigma = \sqrt{\frac{\sum x^2}{N} - \mu^2}$$

These statistics were saved and reused during training, validation and testing to ensure distributional consistency.

3) Standardization (Z-Score Normalization): Each input channel was standardized using:

$$x' = \frac{x - \mu}{\sigma + 10^{-6}}$$

This stabilizes training, accelerates convergence [12] and ensures that all spectral bands contribute equally.

B. Data Augmentation

Because the training dataset is relatively small (1458 patches), data augmentation is crucial for preventing overfitting and improving generalization. We apply stochastic, on-the-fly augmentations only to the training set:

- **Random Horizontal Flip** (probability $p = 0.5$)
- **Random Vertical Flip** (probability $p = 0.5$)
- **Random 90° Rotation** (randomly applying 90°, 180° or 270° with $p = 0.5$)
- **Random Color Jitter** applied only to the RGB bands (random perturbations in brightness, contrast, saturation and hue)
- **Random Gaussian Blur** (kernel size 3×3, probability $p = 0.3$)

IV. Model Selection

To identify the most effective architecture for multispectral wildfire detection, we evaluate a diverse set of deep learning frameworks. Our selection ranges from the standard convolutional baseline to state-of-the-art self-configuring networks and Transformer-based models capable of capturing long-range dependencies.

A. Architectures

1) Modified U-Net: We utilize the U-Net [6] as our primary baseline and apply a lightly modified version of it. To better handle the high-dimensional Sentinel-2 inputs, we make several small improvements over the implementation in baseline [5]: we replace Conv–ReLU blocks with Conv–BN–ReLU for more stable training on multispectral data, add Dropout [13] to reduce overfitting, and switch to an epoch-based training setup with Cosine Annealing learning-rate scheduling.

2) nnU-Net: We further evaluate a self-configuring U-Net obtained through automated architectural optimization. Instead of relying on manual design, the model adjusts its depth, feature width, and patch size based on dataset properties such as spatial resolution and input channels. The resulting network adopts an eight-stage encoder–decoder with double 3×3 convolutions per level, followed by Instance Normalization and LeakyReLU. Downsampling uses strided convolutions, while the decoder mirrors this with upsampling, skip connections, and identical convolutional blocks, ending with a 1×1 segmentation layer.

Unlike BatchNorm–ReLU variants, the InstanceNorm–LeakyReLU configuration improves stability with small batch sizes. Since preprocessing, patch sampling, and batch size are jointly selected with the architecture, the system forms a fully data-driven pipeline that removes the need for manual tuning.

3) SegFormer: We additionally evaluate a transformer-based segmentation approach by using which leverages a lightweight hierarchical transformer backbone (NVIDIA MiT-B0) paired with an MLP decoder that directly aggregates multi-scale features and upsamples logits to the original resolution. To support multispectral wildfire detection, we modify the input projection layer to process all 13 band channels and constrain the model to produce a binary fire mask.

We consider three versions of this model. V03-1 represents the baseline configuration, trained using a BCE+DICE loss with class imbalance handled at the sample level through WeightedRandomSampler, which oversamples tiles containing active fires. V03-3 uses the same architecture but introduces pixel-level spatial weighting, assigning a higher loss to coordinates where fire occurrences are more frequent in the dataset combined with stronger regularization. V03-4 further restricts the model to only four spectral bands and disables colour jitter, serving as a feature-selection variant that tests whether reduced channel input improves generalization.

4) DualStream SegFormer: We further introduce a DualStream SegFormer that explicitly separates high-signal and contextual information in Sentinel-2 imagery. The model employs two independent MiT-B0 encoders: an expert stream dedicated to four fire-sensitive bands (B12, B8, B4 and the aerosol band) and a context stream processing the remaining nine channels. Features from both encoders are extracted at four scales and concatenated channel-wise and pass to a standard SegFormer MLP head for the prediction. The network is trained with a BCE+Dice loss and a spatial pixel-weight map that emphasizes locations with historically frequent fires, using a low learning rate (1×10^{-5}) and strong weight decay (1×10^{-2}).

5) MixedTrans-CNN: We evaluate a hybrid CNN–Transformer segmentation model (CNNTransformerUNet) that combines convolutional feature extraction with transformer-based contextual reasoning. Long-range wildfire patterns are further enhanced through an ASPP module before decoding with a U-Net–style upsampling path and skip connections. The model consumes all 13 Sentinel-2 bands and outputs a binary fire mask.

We examine three variants. V01-3 uses the full 13-band input and is trained with BCE–Focal Tversky loss plus oversampling and class weighting. V01-3-2 keeps the architecture unchanged but restricts input to four high-signal bands (B12, B8, B4, aerosol) to isolate feature-selection effects. V01-5 builds on this reduced-band setup, adding dropout to decoder blocks and switching to BCE+DICE with a reduced learning rate for improved regularization.

B. Test Time Augmentation

To further evaluate the robustness of our trained models, we performed inference on the test set both with and without Test Time Augmentation (TTA). TTA enhances predictive stability by aggregating predictions across multiple transformed views of the same input image. This process aims to reduce prediction variance and suppress noise, effectively increasing model confidence on unseen data. Our TTA strategy generates six distinct views for every test input sample x:

1) The original identity image
2) A horizontal flip
3) A vertical flip
4) A rotation of 90°
5) A rotation of 180°
6) A rotation of 270°

During inference, the model predicts the segmentation mask for each of these six views. We then apply the inverse transformation to each output mask to realign them with the original image geometry. The final prediction is obtained by computing the arithmetic mean of these six aligned probability maps. The quantitative impact of this strategy is detailed in Section V and Table III.

V. Experimental Results

Table III details the quantitive performance of the evaluated models. We report precision, recall and F1-score to align with the metrics used in the original study. Additionally, we extend the evaluation by including also the Intersection over Union (IoU) score to provide a more rigorous assessment of segmentation overlap. Furthermore, the primary values in Table III presents the performance of the models using Test Time Augmentation (TTA) and where applicable, the subscript indicates the absolute change (Δ) compared to standard inference (No TTA). An upward arrow ($\uparrow$) indicates that TTA improved the metric, while a downward arrow ($\downarrow$) indicates a decline.

A. Baseline Replication

Since the pre-trained weights from the original study were not publicly available we trained a reproduction model using only the original code provided by the authors that serves as our comparative baseline. As shown in the first section of Table III our replication of the U-Net (SWIR+Aer) achieved an F1-score of 0.2442, while slightly lower than the reported 0.2810 it still provides a verifiable standard against which our proposed improvements are measured.

B. Architectural Improvements and Input Strategies

The most significant finding of our experiment is the superior performance of the nnU-Net framework. Regardless of the input strategy, the nnU-Net–generated U-Net variants substantially outperformed both the original baseline and our reproduction. Furthermore, the original Sen2Fire study concluded that a subset of bands (SWIR + Aerosol) yielded better performance than using all available bands, attributing this to information redundancy. However, our results with nnU-Net challenge this conclusion, since the configuration trained on all 13 bands achieved the highest overall performance across all experiments. With an F1-score of 0.3675 and IoU of 0.2250, this represents a substantial improvement over the original paper's best reported F1-score (0.2810). This suggests that more robust systems—such as nnU-Net, which automatically selects the most effective U-Net design for the dataset—are capable of fully leveraging the complete spectral information when all bands are used.

C. Transformer-base approaches

We also explored transformer-based architectures such as SegFormer and MixedTransformer variants. While these models showed promise in recall (indicating a strong ability to detect positive fire pixels) they generally suffered from lower precision compared to the nnU-Net. As an example, the SegFormer (Base) achieved a recall of 0.2580 but struggled to maintain the precision levels necessary for a balanced F1-score.

D. Impact of Test Time Augmentation

The integration of Test Time Augmentation proved beneficial for the majority of configurations. As indicated by the subscripts in Table III, TTA consistently improved the IoU and F1-scores for the top-performing models. For the best-performing model (nnU-Net with All 13 bands), TTA contributed a gain of 0.0025 in F1-score and 0.0020 in IoU. While TTA yields marginal performance improvements, its practical utility is constrained by the computational cost since the incremental gains in F1-score and IoU must be weighed against an approximate six-fold increase in inference latency, making it less viable for real-time or resource-constrained applications.

VI. Conclusion and Future Work

In this study, we evaluated and improved upon the baselines for the Sen2Fire wildfire detection dataset. Our primary contribution is the establishment of a new state-of-the-art performance benchmark. By employing the nnU-Net framework, we achieved an F1-score of 0.3675, significantly outperforming the original baseline of 0.2810. Contrary to the findings of the original study, which suggested that restricting the input to specific band subsets was necessary to maximize performance, our experiments demonstrate that advanced architectures can effectively leverage the full spectral information. We observed that our best-performing models utilized all 13 available bands, indicating that robust deep learning frameworks can handle the potential redundancy or noise in additional bands without requiring manual feature selection.

A. Future Works

This work extends the Sen2Fire dataset and establishes a stronger performance baseline for wildfire detection. We aim for it to support future research in developing and benchmarking more advanced models. A key limitation of the dataset is its narrow geographical and temporal coverage—restricted to four regions in Australia during the 2019–2020 bushfire season. Future efforts should expand it to more recent events and diverse global regions to improve model generalization. Additionally, since some of our best models rely on all available bands, future research should explore approaches that remain effective across satellites with different spectral configurations.

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APPENDIX A
APPENDIX

TABLE I
RATIO OF FIRE TO NON-FIRE SAMPLES ACROSS DATASET SPLITS

Split	Fire (%)	Non-fire (%)
Training Set	2.5	97.5
Validation Set	3.6	96.4
Test Set	5.8	94.2

TABLE II
BAND INFORMATION IN THE SEN2FIRE DATASET

Band	Description	Wavelength (µm)	Resolution (m)
B1	Coastal aerosol	0.443	60
B2	Blue	0.490	10
B3	Green	0.560	10
B4	Red	0.665	10
B5–B7	Red edge	0.705–0.783	20
B8	NIR	0.842	10
B9	Red edge	0.865	20
B10	Water vapour	0.945	60
B11	SWIR	1.610	20
B12	SWIR	2.190	20
B13	Aerosol index	—	1113

TABLE III
EXPERIMENTAL RESULTS AND IMPACT OF TEST TIME AUGMENTATION (TTA) ON MODEL PERFORMANCE. MAIN VALUES DENOTE RESULTS **WITH TTA**. SUBSCRIPTS DENOTE THE CHANGE COMPARED TO INFERENCE WITHOUT TTA.

Model Architecture	Input	Precision (TTA)	Recall (TTA)	F1-Score (TTA)	IoU (TTA)
Baselines (Original & Reproduction)					
U-Net (Original Paper)	SWIR+Aer	0.3970	0.2180	0.2810	0.1637
U-Net (Reproduction)	SWIR+Aer	$0.3354_{\uparrow 0.0310}$	$0.1920_{\uparrow 0.0044}$	$0.2442_{\uparrow 0.0120}$	$0.1391_{\uparrow 0.0078}$
U-Net & nnU-Net Variants					
U-Net	All (13)	$0.3192_{\uparrow 0.0019}$	$0.2317_{\uparrow 0.0102}$	$0.2685_{\uparrow 0.0076}$	$0.1551_{\uparrow 0.0051}$
U-Net	SWIR+Aer	$0.4059_{\downarrow 0.0077}$	$0.2643_{\uparrow 0.0079}$	$0.3201_{\uparrow 0.0035}$	$0.1906_{\uparrow 0.0026}$
nnU-Net	All (13)	$0.3747_{\uparrow 0.0064}$	$0.3606_{\downarrow 0.0008}$	$0.3675_{\uparrow 0.0025}$	$0.2250_{\uparrow 0.0020}$
nnU-Net	SWIR+Aer	$0.4199_{\uparrow 0.0043}$	$0.3166_{\downarrow 0.0014}$	$0.3610_{\uparrow 0.0007}$	$0.2200_{\uparrow 0.0010}$
Dual Stream Architecture					
DualStream SegFormer (v03_5)	Split (4+9)	$0.2725_{\downarrow 0.0082}$	$0.3103_{\uparrow 0.0155}$	$0.2356_{\uparrow 0.0018}$	$0.1511_{\uparrow 0.0015}$
MixedTransformer Variants					
MixedTrans-CNN (v01_3)	All (13)	$0.2507_{\uparrow 0.0081}$	$0.2869_{\uparrow 0.0030}$	$0.2218_{\uparrow 0.0070}$	$0.1510_{\uparrow 0.0032}$
MixedTrans-CNN (v01_3_2)	SWIR+Aer	$0.2150_{\uparrow 0.0009}$	$0.3611_{\downarrow 0.0004}$	$0.2103_{\uparrow 0.0004}$	$0.1396_{\uparrow 0.0002}$
MixedTrans-CNN + Drop (v01_5)	SWIR+Aer	$0.2334_{=0.0000}$	$0.2680_{\downarrow 0.0063}$	$0.2139_{\downarrow 0.0040}$	$0.1447_{\downarrow 0.0030}$
SegFormer Variants					
SegFormer Base (v03_1)	All (13)	$0.2828_{\uparrow 0.0213}$	$0.2580_{\downarrow 0.0105}$	$0.2440_{\downarrow 0.0002}$	$0.1493_{\uparrow 0.0007}$
SegFormer + Spatial (v03_3)	All (13)	$0.2397_{\uparrow 0.0136}$	$0.3639_{\uparrow 0.0056}$	$0.2613_{\uparrow 0.0073}$	$0.1619_{\uparrow 0.0056}$
SegFormer + Spatial (v03_4)	SWIR+Aer	$0.2387_{\downarrow 0.0483}$	$0.3204_{\downarrow 0.0310}$	$0.2647_{\downarrow 0.0178}$	$0.1656_{\downarrow 0.0124}$
Naive Comparison					
Simple CNN	All (13)	$0.0734_{=0.0000}$	$0.6966_{\downarrow 0.0003}$	$0.1329_{=0.0000}$	$0.0712_{=0.0000}$